Name:
Neptun code:

Second midterm test
Quantum Physics
BME, physicist-engineer, 9th December

Only a calculator and pens may be used during the test. Mobile phones, tablets, laptops, etc. may not be used as calculators. The exercises must be solved individually. Start: 12:15 Submission: 13:45

You must write on the test paper with a blue pen. Write the final result in the designated box. Once a result has been written in the box, it may not be changed.

If needed, pages 3 and 4 may be used for detailed calculations, extra papers may be asked from the teacher.

All numerical solutions are accepted with a relative error up to 10%.

USEFUL CONSTANTS AND FORMULAS

- reduced Planck constant: $\hbar = 1.034 \times 10^{-34}\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s} = 0.66\ \mu\mathrm{eV}\cdot\mathrm{ns}$.
- elementary charge: $e = 1.6 \times 10^{-19}$ C.
- Boltzmann constant: $k_\mathrm{B} = 1.38 \times 10^{-23}$ J/K $= 86.2 \times\ \mu$eV/K.
- mass of electron $m = 9.1 \times 10^{-31}\,\mathrm{kg}$.
- speed of light: $c = 3.0 \times 10^8\,\mathrm{m/s}$.
- vacuum permittivity: $\epsilon_0 = 8.85 \times 10^{-12}\,\mathrm{A^2 s^4 kg^{-1} m^{-3}}$.
- electron volt: 1 eV $= 1.6 \times 10^{-19}$ J.
- spontaneous emission rate: $B_\mathrm{sp} = \frac{\omega^3 d^2}{3\pi\hbar\epsilon_0 c^3}$.

EXERCISES

1. *Measurement of a local variable.* Consider that Charlie prepared a pair of spins in the entangled state:

   $$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

   Then, he sent the first spin to Alice and the second spin to Bob and Alice measured the polarization of the spin in the x basis (σ_x) to be $+1$. What is the post-measurement state after the local measurement of Alice?

   $|\psi_{+1}\rangle = |{+}{-}\rangle = (|00\rangle + |10\rangle - |01\rangle - |11\rangle)/2$

2. *Correlation in Bell states.* Consider the composite Bell state

   $$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

   Calculate the expectation value of the $\sigma_y \otimes \sigma_y$ observable.

   $\langle \sigma_y \otimes \sigma_y \rangle = -1$

3. *Bell's inequality* Assume that Q, R, S and T are classical observables, with outcomes ± 1. Calculate the upper bound:

   $$|\,\langle QS\rangle + \langle RS\rangle + \langle QT\rangle - \langle RT\rangle\,| \le c,$$

   $c = 2$

4. *Violation of Bell's inequality.* The correlation matrix T is defined as

   $$T_{\alpha\beta} = \langle \sigma_\alpha \otimes \sigma_\beta \rangle\,.$$

   For the singlet state $|\Phi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ the correlation matrix is $T_{|\Phi^-\rangle} = -I$. Calculate the combination of expectation values for the observables in the state $|\Phi^-\rangle$:

   $$E = \langle Q \otimes S\rangle + \langle R \otimes S\rangle + \langle Q \otimes T\rangle - \langle R \otimes T\rangle\,,$$

   where

   $$Q = X,\ R = Z,\ S = \frac{X+Z}{\sqrt{2}},\ T = \frac{X-Z}{\sqrt{2}}.$$

   $E = -2\sqrt{2}$

5. *Locality loophole.* Alice and Bob would like to show that the world is quantum by showing the violation of Bell's inequality. The lab of Alice is located at the roof of BME F building, the lab of Bob is at the top of the Gellért hill in the Citadel. The distance between the two labs is $d \approx 1$ km. Calculate the time constraint for the measurement of Alice and Bob to ensure that communication is forbidden by relativity theory.

   $t = 3.33\ \mu$s

6. *Dimension of the bosonic subspace.* Consider 3 bosons of the same type, with one-particle Hilbert space $\mathbb{C}^4$. Calculate the dimension of the bosonic subspace.

   $d_\mathrm{B} = 20$

7. *Dimension of the fermionic subspace.* Consider 3 fermions of the same type, with one-particle Hilbert space $\mathbb{C}^4$. Calculate the dimension of the fermionic subspace.

   $d_\mathrm{F} = 4$

8. *Bosonic two-particle Hamiltonian.* Consider 2 bosons of the same type, in a double quantum dot, with Hamiltonian

$$H_1 = \frac{\varepsilon}{2}\sigma_z - t\sigma_x.$$

Calculate the non-interacting two-particle Hamiltonian, in the symmetric basis constructed from the one-particle basis states:

$$|L\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |R\rangle = \begin{pmatrix}0\\1\end{pmatrix},$$

ordered as $|2,0\rangle, |1,1\rangle, |0,2\rangle$, where the bosonic states are labeled by the dot occupation.

$$H_{2,\mathrm{B}} = \begin{pmatrix} \epsilon & -t\sqrt{2} & 0 \\ -t\sqrt{2} & 0 & -t\sqrt{2} \\ 0 & -t\sqrt{2} & -\epsilon \end{pmatrix}$$

9. *Polarization angle dependence of transition rates.* Consider a two-level atom that is irradiated by incoherent polarized light. The transition rate is measured and depends on the polarization angle of the light with respect to the z axis as

$$\Gamma(\theta) = \Gamma_0 \cos\theta.$$

Calculate the transition rate when the atom is illuminated with incoherent unpolarized light, but with same intensity. You may use that

$$\int_0^\pi \sin\theta \cos^2\theta \mathrm{d}\theta = \frac{2}{3}.$$

$$\overline{\Gamma} = \Gamma_0/3$$

10. *Homogeneous field approximation.* Consider a two-level atom with a gap of $E = 1$ eV. Calculate the ratio between the Bohr radius and the dominant wavelength of the photons inducing absorption and stimulated emission.

$$\tfrac{\lambda}{a_0} = 23.4 \times 10^3$$

11. *Hopping amplitude from discretization.* Consider the real space, 1D Hamiltonian of the harmonic oscillator:

$$H = \frac{p^2}{2m^*} + \frac{1}{2}m^*\omega^2 x^2.$$

After discretization with central finite differences of lowest order, one gets a tight-binding model of the form:

$$H_{\mathrm{tb}} = \sum_{n=-\infty}^{\infty} t\big(|n\rangle\langle n+1| + |n+1\rangle\langle n|\big) + \epsilon_n |n\rangle\langle n|.$$

Express the parameters of the discrete model t and ϵ_n with the parameters of the harmonic oscillator, and the lattice constant a.

$$t = -\tfrac{\hbar^2}{2ma},\ \epsilon_n = \tfrac{\hbar^2}{m^* a} + \tfrac{m^*\omega^2 a^2}{2} n^2$$

12. *Probability current.* Consider the wavepacket

$$\psi(x) = \sqrt[4]{\frac{1}{\pi a^2}} e^{-\frac{x^2}{2a^2}} e^{-ikx}.$$

Calculate the probability current.

$$j = -\tfrac{\hbar k}{m}|\psi(x)|^2$$

13. *Discrete probability current.* Consider a three site tight-binding model of the form

$$H = t(|1\rangle\langle 2| + |2\rangle\langle 1| + |2\rangle\langle 3| + |3\rangle\langle 2| + |3\rangle\langle 1| + |1\rangle\langle 3|),$$

or in matrix form

$$H = t\begin{pmatrix}0&1&1\\1&0&1\\1&1&0\end{pmatrix}.$$

Calculate the particle current from site 1 to site 2 in the state

$$|\psi\rangle = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\ e^{i\frac{2\pi}{3}}\\ e^{-i\frac{2\pi}{3}}\end{pmatrix}.$$

$$j = \tfrac{2t}{3\hbar}\sin(\tfrac{2\pi}{3}) = \tfrac{t}{\sqrt{3}\hbar}$$

14. *Discretization of the electric field.* Consider a particle in a 2D square box potential. After discretization on an $N \times N$ square lattice using central finite differences, the tight-binding model with lattice constant a reads as

$$H_{\mathrm{tb}} = \sum_{n=0}^{N-1}\sum_{l=0}^{N-1} t_{n,l}\big(|n,l\rangle\langle n+1,l| + |n,l\rangle\langle n,l+1| + |n+1,l\rangle\langle n,l| + |n,l+1\rangle\langle n,l|\big) + \epsilon_{n,l}|n\rangle\langle n|.$$

Switching on a a homogeneous electric field $\mathbf{E} = (E_x, E_y)$ does not modify the form of the tight-binding Hamiltonian. Calculate the contributions to the parameters $t^E_{n,l}$ and $\epsilon^E_{n,l}$ due to the electric field.

$$\epsilon^E_{n,l} = ea(nE_x + lE_y),\ t^E_{n,l} = 0$$

15. *Error of finite difference.* Consider a continuous function $f(x)$. The second derivative of the function can be approximated as

$$f''(x) \approx \frac{f(x+a) + f(x-a) - 2f(x)}{a^2}.$$

Taylor expand the finite difference up to fourth order and find the α and β coefficients of the result defined as

$$\frac{f(x+a)+f(x-a)-2f(x)}{a^2} = f''(x) + \alpha a f'''(x) + \beta a^2 f''''(x).$$

$\alpha = 0,\ \beta = 1/12$

16. *Thermal state of a gas of two-level atoms.* A two-level atom can absorb and emit $\lambda = 3.9$ μm wavelength radiation. Let us take a small box with 10^{23} such atom in it at room temperature $T = 20$ °C. What is the expectation value of the number of atoms in the excited state?

$n_\mathrm{e} = 3.4 \times 10^{17}$

17. *Population inversion.* Consider a three-level atomic system with populations p_1, p_2, p_3 (levels 1 = ground, 2 = upper lasing level, 3 = pump level). The transitions and rates are:

    - Optical pumping: $1 \overset{R}{\longleftrightarrow} 3$ with rate $R =$ 3 MHz.
    - Fast nonradiative decay: $3 \xrightarrow{\gamma_{32}} 2$ with rate $\gamma_{32} = 2$ GHz.
    - Radiative (or other) decay: $2 \xrightarrow{\gamma_{21}} 1$ with rate $\gamma_{21} = 1$ kHz.

    Calculate the ratio p_2/p_1 in the steady state.

$\frac{p_2}{p_1} \simeq \frac{R}{\gamma_{21}} = 3000$

18. *Photon flux from a laser.* Compute the number of photons emitted per second from a helium-neon gas laser with 1 mW power and 600 nm wavelength.

$N = 3 \times 10^{15}$

19. *Resonator of a laser.* Consider a gas laser with 1 mW power and 600 nm wavelength. Assume that the distance between the mirrors of the optical resonator is 15 cm. Estimate how many zeros (nodes) the resonator mode used has in the space between the two mirrors.

$N \simeq \frac{2L}{\lambda} = 500000$

20. *Radiative lifetime.* Consider a two-level atom, with dipole transition matrix element $d = 1\ e \cdot$ Å, where e is the elementary charge, and energy gap $E = 1$ eV. Calculate the radiative lifetime due to spontaneous emission:

$$1/\tau = B_\mathrm{sp}.$$

$\tau = 263$ ns